Statement of Purpose

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My passion for natural language processing (NLP) has driven me to research and publish in this field for three years. Pursuing a related PhD becomes a natural choice for me to dive deeper into the area. I believe NLP technology can (and should) empower humans to achieve a better life, and I am interested in pushing towards trustworthy, positive, and accessible NLP systems with social intelligence. I am currently a first-year PhD student at the Georgia Institute of Technology. As my plans have evolved, I believe your environment is a better fit for my goals. After presenting my motivation, I discuss both my past and future work related to inspecting NLP models/datasets and situating NLP in social contexts, before concluding with broader interests and career goals.

Motivation. The pretraining paradigm produces stunning multipurpose models for natural language understanding (Brown et al., 2020; Devlin et al., 2019; Peters et al., 2018). However, we can not trust these massive black-box models, neither to generalize robustly (Habernal et al., 2018; McCoy et al., 2019) nor to avoid socially undesirable outcomes (Gehman et al., 2020). To establish trust among the parties shown in Figure 1, we need to better understand models as well as corresponding datasets. Closer investigation is needed to characterize their limitations such as fragility under domain shift (Gururangan et al., 2020), inconsistency under perturbations (Li et al., 2020), and lack of commonsense (Zhou et al., 2020c). We must also help models overcome the aforementioned weaknesses. Last but not least, gaining trust requires mutual understanding: I also want models to understand humans better, for which we need to incorporate social context and understanding into models.

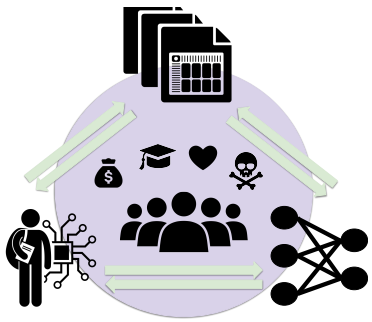


Figure 1: The dynamic relationship among NLP practitioners, datasets, models, and human groups (center). The purple circle represents broader social contexts that connect all the parties but often receive limited attention in building NLP applications. I want to bring humans and relevant social contexts to the center of building and inspecting models and datasets, emphasizing fairness, diversity and positive social impacts.

Inspecting NLP Models and Datasets. Upon seeing the purportedly super-human performance of pretrained language models on various benchmarks in 2018, I became curious about what reasoning abilities they *really* have. That curiosity turned into my paper published in AAAI 2020, where we investigated pretrained language models' commonsense knowledge (Zhou et al., 2020c). Among other experiments, I designed a robustness test by perturbing commonsense test instances to produce minimally edited pairs with different ground truths. In the results, I found that models lacked robust commonsense regardless of their architecture, size, and pre-training corpus. Since these pre-trained language models are brittle on commonsense tasks, what about other tasks and what kinds of perturbations hurt models the most? To address that question, inspired by recent work on *contrast sets* (Gardner et al., 2020), I led a team at UW designing a linguistically informed tool for automatic generation of *contrast sets*, and found that models are far from being robust under linguistic variation, such as changes in English tense. We published this paper at the BlackboxNLP 2020 Workshop (Li et al., 2020).

While at Apple Machine Intelligence, I led two projects. One evaluated gender bias in multilingual models when translating between non-English language pairs. We found that pivoting through English, the common practice of multilingual machine translation, exacerbates gender bias (under review of ACL 2022). The other

ongoing project aims to understand the stereotyping behaviors in multilingual language models and relevant down-streamed applications by collecting datasets that consider 14 high or low-resourced languages with the help of annotators worldwide.

If we compare NLP models to cars, I see the datasets behind them as the "oil". *In the future* 🕒, I want to further investigate the dynamics between models and datasets. This includes but is not limited to: 💡$_1$ Dataset curation for promoting better models, especially positive societal impacts. For example, few-shot learning requires significantly fewer data to train a model, allowing us to curate the training set carefully. To what degree can we control models' behavior (e.g., debias toxic-language classifier) by controlling the small training set? 💡$_2$ Using the models' behavior alone with statistical tools to help us understand the datasets better (Swayamdipta et al., 2020; Ethayarajh et al., 2021), focusing on characterizing data points that lead to socially undesirable issues.

Situating NLP in Social Contexts. Since human language is intertwined with complex social processes, I believe NLP systems must understand the social context in order to build trust between users and systems. For example, toxic language detection systems suffer from various kinds of biases, which risk suppressing minority voices and further exacerbate social inequality. This inherently limits trust in those systems. In my work at EACL 2021 (Zhou et al., 2021), Maarten Sap, Swabha Swayamdipta and I investigated automatically debiasing toxic language detection systems. I comprehensively experimented with debiasing algorithms from both the dataset and model perspectives while considering scenarios where the source of bias is both known and unknown. We found that current debiasing algorithms fail in the task of toxic language detection, and we demonstrated that the main reason is annotation bias in the dataset itself. It is interesting to see how texts encode humans' beliefs and attitudes. I advised a student at UW to investigate ways to infer those personal attributes from dialogues (Wang et al., 2021), which is an important step towards more personalized NLP applications.

Like people, NLP systems do not live in a vacuum; instead, they are situated in highly complex social environments, and they often encode implicit assumptions. *In the future* 🕒, I hope to enable NLP systems to encode social knowledge. For example, the speakers' and audience's background are important to consider when determining toxicity (Waseem, 2016; Sap et al., 2019). However, current approaches for toxic language detection oversimplify the problem and do not consider the pertinent social features, which is the same case for many other NLP applications such as chatbot, question answering, and summarization. 💡 Can we design models that can navigate through text data and incorporate the characteristics of the speakers and listeners? A promising direction for this question could be entity-centric language modeling (with more attention to the human entities specifically), in which human entities alongside their interactions in the texts are being explicitly represented and updated during the training process. Preliminary investigation on this topic has demonstrated the advantage of modeling entities in general language understanding tasks (Ji et al., 2017; Peters et al., 2019; Févry et al., 2020). Based on these investigations, I plan to design human entity-centric algorithms that can efficiently 'learn' social context through massive raw text data.

Other Interests. More generally, I am broadly interested in NLP and computational linguistics. Specifically, I enjoy developing algorithms that can explain their predictions, as demonstrated in my first-author work appearing at EMNLP 2020 (Zhou et al., 2020b). I love the theory behind NLP algorithms, which I studied in my bachelor thesis and in my ACL SRW workshop 2020 paper (Zhou et al., 2020a). My ultimate goal in pursuing a PhD is to become an investigator who can explore areas of interest as well as share what is learned in the research journey with mentees and the broader academic community.

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